

Circuit Simulator — Project Plan

(Duration: 29 Sep, 2025 – 12 Dec, 2025)

Project Overview

Goal:

Develop a basic circuit simulator that enables users to build, analyze, and simulate electrical circuits visually, showing different analysis results in the form of text and/or graphical representation.

Key Focus Areas:

- UI design and graphical interaction
- Circuit modeling and analysis (using MNA and ODE solvers)
- Integration of external libraries for GUI and computation

Tools and Libraries

Programming Language	C++17
Build System	CMake
Graphics Library	SFML
GUI Framework	ImGui via ImGui-SFML
Differential Equation Solver	Boost.Odeint
Version Control	version.aalto.fi
Documentation	Doxygen
Task Management	version.aalto.fi issue board

Project Scope

Basic Components:

- Resistor, Capacitor, Inductor, Constant Voltage Source

Core Features:

- Graphical UI for circuit creation (add, connect, move, remove components)
- Save and load circuits from files and other
- RMS voltage and current calculations
- Ready-made example circuits

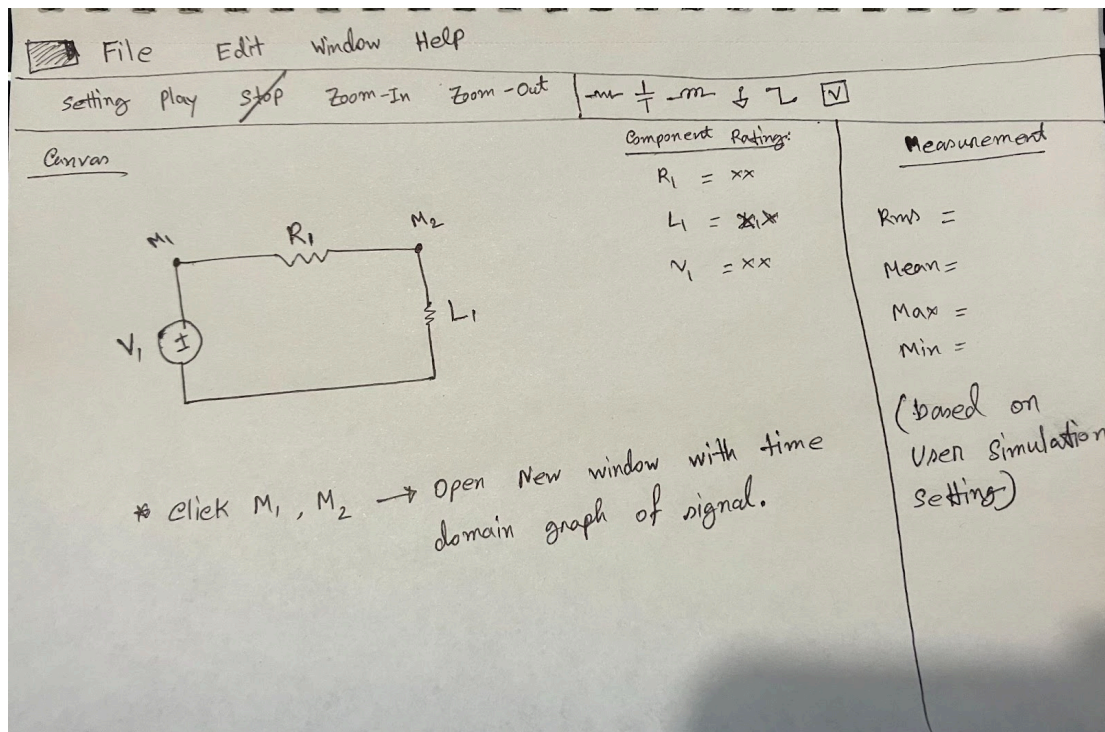
Additional Features:

- Sinusoidal AC voltage source and AC/DC current sources
- Triangle and square waveform sources
- Dependent sources (VCVS, VCCS, CCCS, CCCS)
- Only Time-domain voltage/current graphs

High-Level Architecture

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GUI Layer	
- SFML window & drawing	
- ImGui interface (toolbar, menus, circuit view)	
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Simulation Engine	
- Component classes (R, L, C, sources)	
- Modified Nodal Analysis (MNA) solver	
- ODE solvers (Boost.Odeint) for time-domain	
- Fourier transforms for frequency-domain	
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Data Layer	
- Circuit file I/O (JSON or XML)	
- Ready-made example circuits	
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Circuit Simulator Sketch



Menubar:

File -> New, Open, Save, Close window

Save -> save canvas drawing in png file

Measurements in txt file

Graph in png file (if window opened on canvas)

** Simulation Ran -> all measurements done

-> graph window opened (if selected)

-> Hit save to save in a folder

Edit -> Undo, Redo

Window -> Exit Full Screen, Full Screen

Help -> click to version.aalto.fi Readme page

Toolbar: (All Icon)

Setting -> Selecting different measurement options (RMS, Mean, Max, Min ..)

Choose Option -> Graph (only time domain)

Basic simulation analysis settings*

Play -> Run Simulation and show results and graph

Zoom-in -> not important yet

Zoom-out -> not important yet

Components -> R, L, C, Wire, V, GND

Canvas: Circuit drawing area

Add component -> Drag and drop and move

Delete Component -> Select and hit button

(delete or back space or button from Tool bar X)

Component rating -> At the top right shows all component ratings (may be not ideal)

May be next to each component name.

Rating change(R1, V1 ..) -> Double click on component open new window and fill

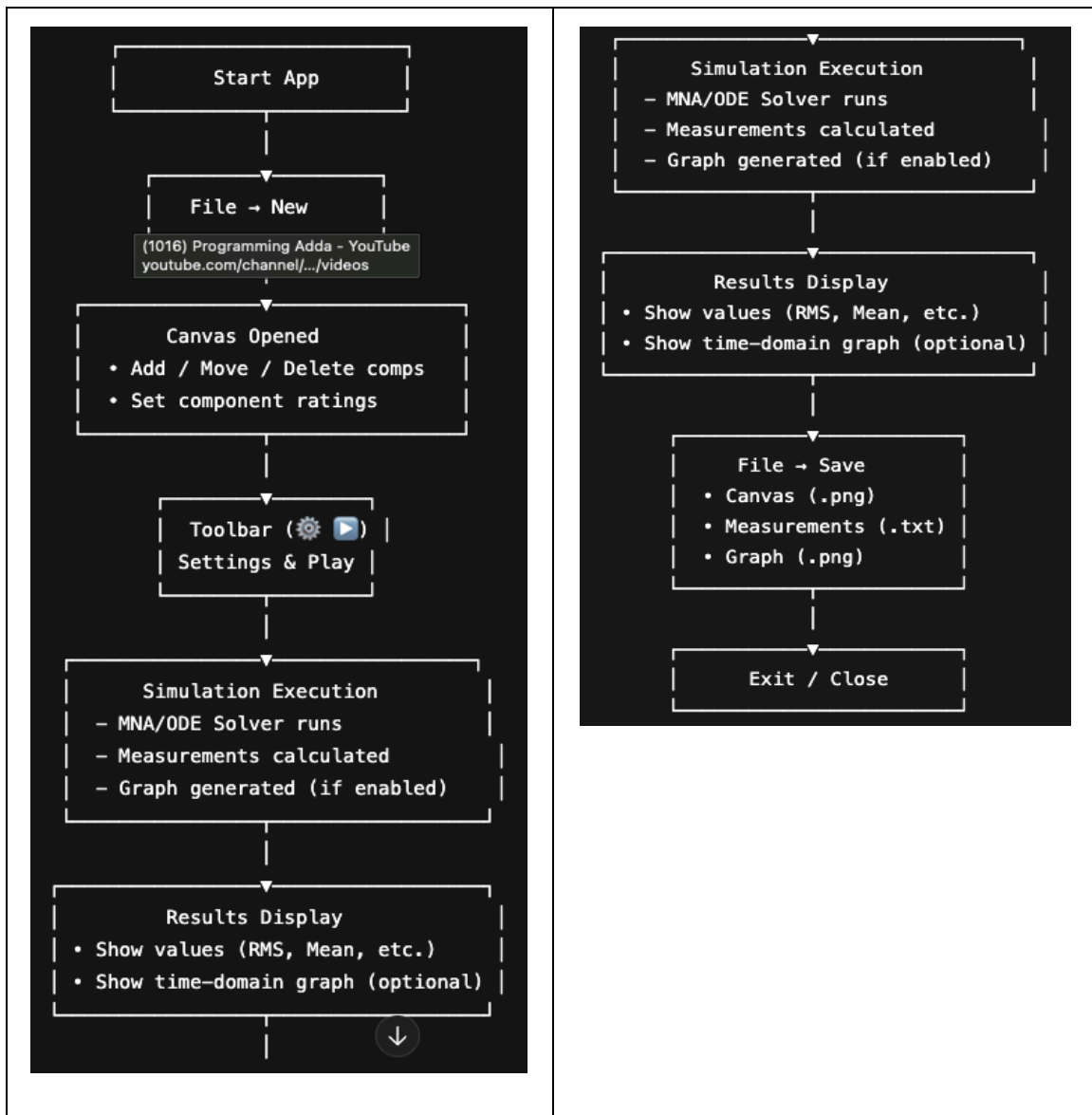
rating.

Measurement -> At right side of canvas but At bottom may look good.

user click M1 and M2 points (need to select two points) to get measurements across a component.

Graph -> if selected in setting. Open new window with time domain graph.

Usage Flow Diagram



Sprint 0 + 1 — Project Setup & Tools Integration

Duration: 29 Sep – 24 Oct

Meetings: 3 × 2 hours (Kick-off, Midpoint, Review)

Goal: Establish project setup, libraries, and foundational structure.

Task	Description	Deliverable
Project planning	Define scope, objectives, and schedule	Project Plan Document
Environment setup	CMake configuration, Git repository	Buildable skeleton
Library integration	Set up SFML, ImGui-SFML, and Odeint	Linked and tested libraries
Class templates	GUI, component, solver structure	UML / class diagram
Documentation	Style guide and setup instructions	README and docs

Sprint 2 — Initial Implementation

Duration: 27 Oct – 7 Nov

Meetings: 2 × 2 hours (Kick-off, Review)

Goal: Implement GUI and basic circuit features.

Task	Description	Deliverable
GUI implementation	Add, move, delete, connect components	Working GUI interactions
Component modeling	Implement R, L, C, voltage source	Core component classes
RMS solver	Calculate RMS values using MNA	Functional solver backend
File I/O	Save and load circuit data	JSON circuit files
Basic circuits	Example: RC and RL circuits	Demonstration templates

Sprint 3 — Feature Completion

Duration: 10 Nov – 21 Nov

Meetings: 2 × 2 hours (Kick-off, Review)

Goal: Add waveform and dependent sources and visualization.

Task	Description	Deliverable
Waveform sources	Implement sinusoidal, triangle, and square sources	Extended source types
Dependent sources	Implement VCVS, VCCS, CCVS, CCCS	Controlled source classes
Graph plotting	Add time- domain graphs	Graphs via ImGui
Testing	Validate main features and components	Test reports

Sprint 4 — Finalization & Documentation

Duration: 24 Nov – 5 Dec

Meetings: 2 × 2 hours (Kick-off, Final Review)

Goal: Final testing, documentation, and demo preparation.

Task	Description	Deliverable
Testing	Verify solver accuracy and GUI stability	Test report
Bug fixes	Identify and resolve issues	Stable final build
Documentation	User manual, developer guide, known issues	Final project documentation
Demo preparation	Prepare demo circuits and presentation	Demo-ready version

Meeting Schedule Overview

Sprint	Duration	No. of Meetings	Meeting Duration	Meeting Purpose
0 + 1	29 Sep – 24 Oct	3	2 hr each	Kick-off, Project Setup & Tools Integration
2	27 Oct – 7 Nov	2	2 hr each	Initial Implementation
3	10 Nov – 21 Nov	2	2 hr each	Feature Completion
4	24 Nov – 5 Dec	2	2 hr each	Finalization & Documentation
Total Meetings: 9 (each 2 hours)				
Total Discussion Time: 18 hours				